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Marie Setaire

Research group of the methods of science education and popularization

Abstract

Cross-modal reorganization refers to the process by which auditory brain regions begin handling non-auditory — mainly visual — information after peripheral hearing loss. Traditionally, this has been seen as an adaptive form of brain plasticity. Growing evidence, however, reveals a more complex picture. In people with normal hearing, audiovisual predictive mechanisms help facilitate speech perception. When hearing is acutely degraded by noise vocoding, the temporal voice areas (TVA) maintain voice-selectivity by drawing on preserved temporal cues. But with prolonged deafness, the TVA undergo structural cross-modal reconfiguration, which produces a lasting visual bias in voice processing that persists even after successful cochlear implant (CI) rehabilitation. This review follows the continuum from normal predictive coding, through the effects of acute degradation, to long-term TVA reorganization. It examines what drives the shift from adaptive to maladaptive reorganization and considers how neuroimaging might one day be used to monitor this process in clinical settings.

Keywords: cross-modal reorganization; temporal voice areas; cochlear implant; predictive coding; vocoding; deafness; brain plasticity; fNIRS; aural preference; audiovisual integration

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Review Article

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Cross-modal reorganization refers to the process by which auditory brain regions begin handling non-auditory — mainly visual — information after peripheral hearing loss. Traditionally, this has been seen as an adaptive form of brain plasticity. Growing evidence, however, reveals a more complex picture. In people with normal hearing, audiovisual predictive mechanisms help facilitate speech perception. When hearing is acutely degraded by noise vocoding, the temporal voice areas (TVA) maintain voice-selectivity by drawing on preserved temporal cues. But with prolonged deafness, the TVA undergo structural cross-modal reconfiguration, which produces a lasting visual bias in voice processing that persists even after successful cochlear implant (CI) rehabilitation. This review follows the continuum from normal predictive coding, through the effects of acute degradation, to long-term TVA reorganization. It examines what drives the shift from adaptive to maladaptive reorganization and considers how neuroimaging might one day be used to monitor this process in clinical settings.

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***Corresponding author:** Marie Setaire

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1. Introduction

The human brain is remarkably flexible, constantly reshaping its functional organization in response to changes in what the senses provide. One of the most studied examples of this plasticity is what happens to the auditory cortex after deafness: regions that normally handle sound start responding to visual, tactile, or other non-auditory signals. Early influential research viewed this reorganization as straightforwardly adaptive — a smart reallocation of "idle" cortical resources toward intact sensory channels (Lee et al., 2001; Sadato et al., 2004). The logic was simple: the more auditory cortex was taken over by cross-modal inputs, the more efficiently the brain was making use of its remaining senses.

Two decades of neuroimaging and psychophysical research have seriously complicated this optimistic picture. How extensively the auditory cortex has been reorganized by visual input before cochlear implantation turns out to be a negative predictor of speech recovery after the implant: the more thoroughly auditory cortex has adapted to visual processing during deafness, the harder it is for that cortex to re-establish auditory specialization once the device is activated (Lee et al., 2007; Doucet et al., 2006; Lazard et al., 2014). What's more, even patients who recover excellent speech intelligibility with a CI show a persistent and abnormal tendency to let visual information dominate voice processing — something that simply does not happen in normally hearing listeners, even under severe acoustic degradation (Barone et al., 2016).

Making sense of this paradox requires looking at audiovisual interactions across a continuum. First, we need to understand how they work normally — as predictive mechanisms in normally hearing listeners (Strelnikov et al., 2015; van Wassenhove et al., 2005). Next, we can examine what happens under acute spectral degradation modeled by vocoding, where TVA voice-selectivity largely holds up through preserved temporal cues (Strelnikov et al., 2011; Belin et al., 2000). Finally, we need to understand what changes with long-term deafness, where structural TVA reorganization leaves behind a functional legacy — visual bias in voice processing — that CI rehabilitation cannot fully undo (Barone et al., 2016).

The review closes by considering what these findings mean for clinical practice and by reviewing the methodological tools most suitable for tracking cross-modal reorganization across the lifespan — including fNIRS protocols developed specifically for use with children (Calmels et al., 2022).

2. Normative Basis: Predictive Coding in Normally Hearing Listeners

2.1. Visual Predictions in Speech Perception

Face-to-face conversation is our most natural form of communication. When someone speaks, their lip movements consistently precede the corresponding sounds by 100–200 ms — and the auditory system makes good use of this timing. Visual speech activates auditory cortex even when there is no sound at all (Calvert et al., 1997), and seeing congruent lip movements shortens the latency of auditory cortical responses (van Wassenhove et al., 2005; Besle et al., 2004). These findings fit naturally into a predictive coding framework, in which the visual channel provides a running forward model of what the auditory system is about to receive (Arnal et al., 2009; Arnal et al., 2011).

Whether similar predictive mechanisms apply to prosodically relevant facial movements — such as eyebrow raises and head nods, which operate on longer timescales of 400–500 ms — had not been tested before. An EEG oddball study using a three-word phrase provided a clear answer (Strelnikov et al., 2015). Standard stimuli presented a neutral audiovisual phrase; deviants used the identical audio but paired it with

a visual prosodic accent (an eyebrow raise) on the middle word, about 500 ms before its expected acoustic position. An MMN-like negativity appeared at 360–400 ms — roughly 200 ms before the acoustically unaccented word arrived. The behavioural correlate was an auditory accent illusion experienced by every participant: the brain "heard" stress that was never acoustically present, generated entirely by the visual signal.

2.2. Neural Architecture of the Audiovisual Predictive Cycle

Spatiotemporal analysis of the MMN-like component traced its propagation from occipito-parietal to fronto-temporal sites and back to occipital regions — an occipito-temporo-frontal loop that completed itself before acoustic target onset. Source analysis identified generators in left frontal and left posterior temporal cortex, in line with the dual-pathway model of audiovisual speech prediction (Arnal et al., 2009). That model describes a fast, direct cortico-cortical route that carries visual motion parameters to auditory cortex, alongside a slower feedback pathway that signals prediction errors. The posterior temporal generator is also consistent with the proposed role of this region in predicting upcoming auditory phonemes from visual context (Kilian-Hutten et al., 2011).

Importantly, the MMN-like response disappeared entirely when the same facial movements were shown without any sound. This rules out a purely visual interpretation and confirms that the response reflects anticipatory audiovisual binding. This normative mechanism provides the baseline against which we must measure changes under acute degradation and chronic deafness. It operates specifically at the prosodic level — which is also the level most severely disrupted by CI processing, since CIs discard the spectral fine structure needed for perceiving fundamental frequency and voice quality (Marx et al., 2015).

3. Acute Degradation Effects: Vocoding and the First Stage of Adaptation

3.1. Vocoding as a CI Model: Separating Acoustic and Linguistic Effects

Noise vocoding mimics the spectral degradation produced by CI processing. It breaks the signal into frequency bands and retains temporal amplitude envelopes while discarding fine spectral structure (Shannon et al., 1995; Lorenzi et al., 2006). Temporal envelope information alone is enough to support voice/non-voice classification, making vocoding a useful tool for separating acoustic from linguistic contributions to cortical responses. Voice/non-voice discrimination holds up much better under vocoding than lexical intelligibility does: at four to eight channels, voice discrimination stays near ceiling while intelligibility is already badly impaired (Massida et al., 2011). This behavioural dissociation implies that there should be a corresponding neural dissociation at different levels of the auditory hierarchy.

A factorial fMRI design crossing stimulus category (intelligible speech, IS; unintelligible speech in an unfamiliar language, US; environmental sounds, ES) with five vocoder degradation levels confirmed this prediction (Strelnikov et al., 2011). Regression analysis revealed a significant correlation between BOLD signal and degradation level in a large bilateral STG/STS cluster, and this effect was comparable for IS and US — yielding no significant whole-brain Category × Degradation interaction. The main driver of superior temporal deactivation under vocoding is therefore acoustic rather than linguistic. A significant intelligibility-specific interaction did appear in the posterior STG, consistent with its role at the phonological–semantic interface (Hickok & Poeppel, 2007; Obleser et al., 2008).

3.2. TVA Voice-Selectivity: Resilience Through Temporal Cues

The temporal voice areas (TVA), distributed along the bilateral STS (Belin et al., 2000; Pernet et al., 2015), responded to vocoder degradation in a qualitatively different way. Region-of-interest analysis using an independent TVA localizer found no significant correlation between TVA activity and degradation level. When degraded stimuli were directly compared to natural ones, the drop in right STS activity was significantly smaller for speech (IS + US combined) than for environmental sounds, showing that the TVA retained a relative advantage for vocal over non-vocal sounds even under two-channel vocoding (Strelnikov

et al., 2011). This resilience comes from the TVA's ability to draw on preserved temporal cues: both intelligible and unintelligible speech share a quasi-periodic syllabic envelope structure and characteristic amplitude modulation patterns that set vocal sounds apart from non-vocal ones, regardless of spectral content (Chandrasekaran et al., 2009).

This resilience sets an important baseline. Acute vocoding — a reversible form of spectral impoverishment — leaves TVA specialization largely intact. Under these conditions, viewing a face does not pull auditory voice categorization off course: normally hearing listeners categorising voice gender under two-channel vocoding showed no measurable shift in their judgements when simultaneously shown an incongruent face (Barone et al., 2016). Acoustic degradation alone, no matter how severe, is not enough to produce the kind of visual takeover of voice processing seen in CI users with reorganized auditory cortex. This contrast between acute and chronic effects is central to the argument developed here.

4. Long-Term TVA Reorganization: The Maladaptive Legacy of Deafness

4.1. Cross-Modal Takeover of the TVA

With prolonged peripheral deafness, the TVA are cut off from the auditory input they depend on. Converging neuroimaging findings show that the STS, as part of the TVA network, progressively takes on visual processing: it responds to visual sign language in early-deaf signers (Sadato et al., 2004), to visual speech in post-lingual CI patients (Rouger et al., 2012), and to static face images in CI users — with the cross-modal response amplitude correlating with face memory performance (Stropahl et al., 2015). Together, these findings suggest that face processing has been genuinely absorbed into the functional repertoire of auditory temporal cortex.

The clinical significance of this reorganisation becomes clear when we look at CI outcomes. Greater pre-operative visual activation of auditory cortex predicts poorer speech recovery after implantation (Lee et al., 2007; Doucet et al., 2006): a cortex that has adapted well to visual processing has, in doing so, become less able to handle the auditory information the CI will later deliver. This negative relationship is strong evidence for the maladaptive side of reorganisation.

4.2. Visual Bias in Voice Processing: Evidence from Experienced CI Users

The functional impact of TVA reorganization on voice processing was assessed directly in a study of fourteen experienced CI patients, selected specifically for high speech intelligibility recovery (mean word recognition 86% correct). This design controls for the inverse effectiveness account by equating unisensory performance across groups (Barone et al., 2016). Participants categorised voice gender along a morphed male-to-female voice continuum while simultaneously viewing congruent or incongruent faces. A visual impact index (VIx) quantified how much face presentation shifted their auditory judgements.

In normally hearing listeners — whether tested under normal hearing conditions or two-channel vocoding — face presentation had no measurable effect on voice categorisation. In CI patients, however, viewing an incongruent face significantly shifted psychometric functions toward the face's gender. The VIx was significantly more negative in patients than in either normally hearing group, the effect held after excluding the three most extreme patients, and reaction time data provided further evidence of heightened face sensitivity (Barone et al., 2016). Crucially, VIx showed no significant correlation with deafness duration, CI experience, or speech comprehension score. The visual bias reflects a stable functional property of the reorganised TVA — not a compensatory strategy that fades as auditory recovery improves.

4.3. Proposed Mechanism: Strengthened Face–Voice Cortical Connectivity

In normally hearing individuals, face-selective regions — most notably the fusiform face area (FFA) — are already directly connected to the TVA, both structurally and functionally (Blank et al., 2011; von Kriegstein et al., 2005). This connectivity supports the integration of facial and vocal information when we recognise

people. During deafness, as the TVA begin responding to face stimuli, repeated co-activation of FFA and TVA is thought to strengthen this connectivity through Hebbian plasticity. After CI restoration, TVA activity partially returns to auditory processing, but the strengthened face–TVA connection remains. This creates a structural basis for visual bias: whenever the auditory and visual signals are ambiguous or competing, CI users are more likely than normally hearing listeners to favour the visual channel.

5. Conditions Governing the Adaptive–Maladaptive Transition

5.1. Sensitive Periods and Temporal Dynamics

Animal models of congenital unilateral deafness have established the existence of sensitive periods for binaural and voice-selective cortical organization. Studies in cats show that auditory cortex neurons on the same side as the hearing ear gain disproportionate representation during a critical developmental window following unilateral auditory deprivation — a pattern termed "aural preference syndrome" (Kral et al., 2013). Restoring bilateral auditory input after this window has closed produces far less cortical normalisation than intervening earlier (Kral & O'Donoghue, 2010).

Very similar findings have emerged in children who receive sequential bilateral CIs. When the gap between the two implants exceeded approximately 1.5 years, children showed a persistent abnormal ipsilateral aural preference that did not normalise even with extended bilateral device use; when the gap was shorter than this threshold, both ears produced symmetric contralateral cortical responses (Gordon et al., 2012). Children with early single-sided deafness showed comparable abnormal aural preference, which was only partially reversed by sustained CI use (Lee et al., 2020). Taken together, these findings point to a temporal window — and, by extension, a therapeutic window — within which the adaptive-to-maladaptive transition takes place and within which timely intervention can prevent or at least partially reverse it.

5.2. Domain Specificity: Voice Attributes Versus Speech Intelligibility

The visual bias seen in experienced CI users (Barone et al., 2016) is specific to voice attribute processing — gender, and by extension emotion and identity — rather than speech intelligibility overall. This selectivity is telling. Good intelligibility recovery provides the TVA with sustained, rich auditory input that can partially compete with established visual signals. Voice attributes are different: they depend on spectral fine structure (fundamental frequency, formant patterns, voice quality) that CI processing cannot reliably reproduce. As a result, the acoustic basis for voice attributes remains chronically degraded even in the best CI users, which keeps the visual face channel at an advantage and sustains the structural dominance that formed during deafness.

This has a clear clinical implication. Standard audiometric assessments of CI outcomes focus on speech intelligibility, and so systematically underestimate how persistent the maladaptive TVA reorganization is. Adding tests of voice attribute processing (gender categorisation, emotion recognition) and audiovisual congruence sensitivity would provide more sensitive functional markers of TVA reorganisation status, and these should become part of routine longitudinal follow-up.

5.3. Individual Variability

Visual bias varies considerably between CI users, and this variation does not correlate with clinical parameters such as deafness duration or CI experience (Barone et al., 2016). This suggests that other factors are driving the adaptive-to-maladaptive transition. Possible moderators include the individual’s pre-existing audiovisual network architecture, genetic differences in plasticity-related signalling pathways, and variation in the compensatory strategies developed during the period of deprivation. Identifying these moderators — for instance through pre-operative structural connectivity and resting-state network mapping — could help predict which patients are most likely to develop maladaptive reorganisation and most likely to benefit from targeted audiovisual rehabilitation.

6. Monitoring Cross-Modal Reorganization: Methodological Perspectives

6.1. Neuroimaging Biomarkers in Adults

In adults, fMRI and PET remain the gold standard for measuring cross-modal reorganisation. The pre-operative level of visual STS/STG activation is predictive of CI outcomes and has prompted calls for neuroimaging-based candidacy screening (Lee et al., 2007; Lazard et al., 2014). An important complement to neuroimaging is the use of psychophysical audiovisual paradigms: measures of voice attribute processing and audiovisual congruence sensitivity pick up functional aspects of TVA reorganisation that standard audiometric tools miss (Barone et al., 2016). Longitudinal studies tracking both neuroimaging and psychophysical markers from before implantation through extended follow-up will be needed to map the full trajectory of TVA reorganisation and recovery.

6.2. fNIRS for Pediatric Monitoring

Using fMRI and PET with children is substantially limited by scan duration, movement sensitivity, and radiation exposure. Functional near-infrared spectroscopy (fNIRS) offers a practical alternative: it is non-invasive, tolerates moderate movement, can be used in child-friendly environments, and has shown acceptable reliability for auditory speech responses at the group level (Wiggins et al., 2016). Studies in infants as young as four months have used fNIRS to show left-hemispheric lateralisation of speech-evoked cortical blood flow (Minagawa-Kawai et al., 2011), confirming that it works reliably across the full paediatric age range.

A protocol designed specifically to assess cross-modal reorganisation in children aged 5–16 years with untreated single-sided deafness (SSD) has been developed and registered (Calmels et al., 2022). The paradigm presents environmental sounds, non-speech vocal sounds, and speech under monaural (left or right ear) and binaural conditions, measuring oxy- and deoxyhemoglobin responses over auditory cortex. Two indices derived from these responses capture the reorganisation: the asymmetry index (AI), which quantifies interhemispheric imbalance during monaural stimulation, and the aural preference index (API), which captures the relative influence of each ear on each hemisphere. Based on the aural preference syndrome hypothesis, children with SSD are expected to show an ipsilateral shift in the AI, with the magnitude of the shift correlating with deafness duration and age — tracking the gradual transition from adaptive compensation toward maladaptive structural fixation (Calmels et al., 2022).

6.3. Toward Longitudinal Biomarker Integration

There is a pressing need to integrate cross-modal reorganisation monitoring into longitudinal clinical pathways — spanning diagnosis, the period of deprivation, rehabilitation, and extended follow-up. Cross-modal reorganisation in CI users partially reverses as auditory speech comprehension returns (Rouger et al., 2012; Chen et al., 2016), but the functional legacy in voice attribute processing persists even when intelligibility recovery is excellent (Barone et al., 2016). This dissociation calls for domain-specific monitoring: speech intelligibility alone is not enough. Voice gender and emotion perception, audiovisual congruence sensitivity, and cortical asymmetry indices all need to be tracked across the full rehabilitation trajectory to understand how far TVA recovery actually goes.

7. An Integrative Model: Three Phases on a Single Continuum

The evidence reviewed in sections 2–6 supports a description of auditory cortex cross-modal reorganisation in deafness as a continuous process with three functionally distinguishable phases, separated by dynamic threshold transitions.

Phase I — normative predictive coding. In normally hearing individuals, the TVA are specialised for voice processing, and audiovisual interactions operate via predictive coding mechanisms that pre-activate auditory representations without disrupting their autonomy (Strelnikov et al., 2015; van Wassenhove et al., 2005). Even under acute spectral degradation (vocoding), TVA retain voice-selectivity through temporal

cues, and visual face information does not pull auditory voice categorisation off course (Strelnikov et al., 2011; Barone et al., 2016). No cross-modal structural reorganisation takes place.

Phase II — compensatory cross-modal adaptation. As peripheral deprivation grows or persists, the TVA begin handling visual stimuli. At this stage, reorganisation brings real functional benefits: better lip-reading, improved audiovisual speech integration, and more effective communication in the absence of hearing. The cortical change is partially reversible if auditory input is restored within the sensitive period. The shift from Phase I to Phase II is shaped by how long deprivation lasts, how severe the hearing loss is, the age at onset, and individual differences in plasticity.

Phase III — maladaptive structural reorganisation. The TVA have been structurally reconfigured, and the visual bias in voice attribute processing is now robust and does not reverse with auditory recovery (Barone et al., 2016). The strengthened face–TVA connectivity is a maladaptive legacy that limits the benefits of CI rehabilitation in everything beyond speech intelligibility. A therapeutic window exists between Phases II and III — defined by the sensitive period for voice-selective and binaural cortical organisation — within which early intervention can prevent or at least partially reverse this maladaptive fixation.

Tracking each patient's position on this continuum — using fNIRS asymmetry indices in children (Calmels et al., 2022), fMRI or PET cross-modal activation maps in adults, and psychophysical measures of voice attributes and audiovisual congruence (Barone et al., 2016) — is the essential foundation for personalised, phase-appropriate rehabilitation.

8. Clinical Implications

The three-phase model points to concrete recommendations across the rehabilitation pathway. For screening, pre-operative neuroimaging of visual auditory cortex activation should be used alongside standard audiometric CI candidacy criteria, since the extent of reorganisation predicts the trajectory of auditory recovery. For timing, the sensitive period evidence makes a strong case for early intervention in children with SSD: fitting bilateral devices within approximately 1.5 years of onset can prevent aural preference fixation (Gordon et al., 2012; Kral et al., 2013).

For outcome evaluation, voice attribute processing measures and audiovisual congruence sensitivity should complement standard intelligibility metrics during longitudinal follow-up. For rehabilitation content, the predictive prosodic coding mechanism described in Section 2 (Strelnikov et al., 2015) provides a rationale for developing training protocols that systematically incorporate prosodically informative facial movements. This could offer Phase III patients a compensatory route to better prosodic comprehension, provided they retain the audiovisual predictive circuitry established during normal hearing.

9. Conclusion

Cross-modal reorganisation of the auditory cortex in deafness is neither simply adaptive nor simply maladaptive. It is a dynamic process whose functional consequences depend on when deprivation begins, how long it lasts, and who it happens to. In normally hearing individuals and under conditions of acute degradation, audiovisual interactions operate through adaptive predictive mechanisms that keep TVA specialisation intact. With prolonged deafness, the TVA undergo structural cross-modal reconfiguration, producing a persistent visual bias in voice attribute processing that cannot be fully corrected even after successful CI rehabilitation.

A therapeutic window exists between adaptive compensation and maladaptive fixation. Knowing where each patient sits on this continuum — using neuroimaging, psychophysical, and behavioural markers that are sensitive to TVA reorganisation status — is essential for planning truly individualised rehabilitation. The development of fNIRS-based monitoring for children with single-sided deafness represents a methodologically well-suited step toward tracking this continuum from its earliest and most reversible

stages right through to the end of the rehabilitation trajectory.

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